Chap 1 Command Line Arguments

Date No.

- Make the input & output filename.ext as variables
- When you run the program from the command line, also pass in the input and output files.

command line arguments

argc: (Argument Count)

An integer that holds the total no. of arguments passed to the program, including the program name itself

argv: (Argument Vector)

An array of char* (string pointers) that points to each command line argument.

"encode10"

argv[0] → Always the name / path of the executable program

argv[1] → The first user-provided argument "secret7.inf10"

argv[2] → second "send7.outf10"

Contents of string are entered at the command line.

General command-line syntax

progname arg1 arg2 ... argn

- the name of the executable file
- optional arguments, separated by spaces or tabs
- All arguments are treated as character strings

eg.
```
int main(int argc, char *argv[])
{
    FILE * indata, * outdata;
    indata = fopen (argv[1], "r");
    outdata = fopen (argv[2], "w");
}
```

operations

P1
- Windows: type cmd and press Enter (to open CMD) press Enter
  On Windows 7 and above: press Windows + R, type cmd.
- cd directory_name (change directory)
- cd .. (return to parent directory) multiple times)
- cd directory_name / directory_name / ... (change directories
- cd ../../... (return to parent directories multiple times)
- Tab key (auto-completion) the current drive)
- cd \ (changes the current directory to the root directory of

backflash

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P2 · dir (display contents of the current directory)
· dir directory_name / directory_name /...
(display contents of the specified directory)
· dir /a (display all contents of the current directory, including hidden files)

P3 · cls (clear the screen)
· dir *.* (display all files in the current directory that match the wildcard)
· ↑ key (command history)
· Enter a filename (open/execute the file)
· command /? (view help information) eg. cd /?
· rename ac1-3.exe Convert.exe

1) Relative Path Method
All initial files must be in the same directory (including folders, desktop, etc.)
① ac > cd + directory address (Navigate to the target directory
② type [target file] (Prints the contents of the target file in the command prompt.
③ copy [source file full name (including extension)] [initial file]
(automatically generates the target file in the command prompt)

2) Absolute Path Method
① Right-click to open and copy the file address (this copied address is an absolute path)
② In the command prompt:
"[source file address]" "[file 1 address]"....
③ type [target file] (Prints the content)

Note: if the output file does not exist yet:
· if only the file name (relative path) is written in the program, the operating system will create the output file in the Current Working Directory (CWD) when the program runs.

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Program.

encode.c

```
#include<stdio.h>
int main(int argc, char* argv[])
{
    FILE* indata, * outdata;
    char this1;
    indata = fopen(argv[1], "r");
    outdata= fopen(argv[2], "w");
                :
    fclose(indata); fclose(outdata);
    return 0;
}
```

covert.c

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#define SIZE 10
int main(int argc, char* argv[])
{
    int digit, base, remainder, i;
    char symbol[SIZE];
    if(argc!=3)
    {
        printf("Wrong Format!");
        exit(1);
    }
    if(strlen(argv[2])!=7)
    {
        printf("Base Format Error!");
        exit(1);
    } dereference of address +6 : the seventh character
    if(*(argv[2]+6)>'8')
    {
        printf("Base should be less than 9!\n");
        exit(1);
    }
```

turn into num.

```
base=*(argv[2]+6)-'0';
digit=atoi(argv[1]); turn string into integer
if(digit<0){
    putchar('-');
    digit=-digit;
}
i=0;
do{
    remainder=digit%base;
    symbol[i]=remainder+'0';
    i++;
    digit=digit/base;
}while(digit>0);
do{
    i--;
    putchar(symbol[i]);
}while(i!=0);
putchar('\n');
}
```

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